Pedro Oliveira

07432 654140 | pedro.ddvo@outlook.com | github.com/pedroddvo | Pedro Oliveira

Profile

Mathematics undergraduate specialising in statistical modelling, machine learning and computational methods. Experienced in applying quantitative techniques to real-world datasets through academic and independent projects using Python, R and SQL.

Education

University of Sheffield

BSc in Mathematics (Graduating in July 2026) — September 2023 - July 2026

- Expected classification: First Class (or 2:1)
- Relevant modules: Machine Learning, Operations Research, Time Series, Stochastic Processes and Finance, Statistical Inference and Modelling
- Focus areas: Statistical modelling, data analysis, machine learning, optimisation and computational methods

Technical Skills

Programming Languages	**Python (Pandas, NumPy, Matplotlib, SciPy)**, SQL, R (tidyverse), C++, C
Tools	**Excel**, Git, Github, Visual Studio
Spoken Languages	**English**, Portuguese

Projects

Red vs Grey Squirrel Population Analysis — pedroddvo/GreyVsRed

- Collected and cleaned UK wildlife observation data from NBN Atlas
- Studied relationships between grey squirrel spread and red squirrel population trends
- Performed exploratory data analysis and statistical modelling
- Produced visualisations highlighting regional and temporal trends
- Managed project development using Git and published code and documentation on GitHub

Academic Projects

Statistical Modelling of Paper Helicopter Dynamics (1st Class) — pedroddvo/PaperHelicopter

- Designed and conducted a statistically rigorous experiment investigating factors affecting paper helicopter flight performance
- Analysed data from 150+ experimental flights using R, exploratory data analysis and visualisation techniques
- Applied design of experiments methodology to evaluate the effects of multiple experimental variables
- Developed and interpreted linear models to quantify relationships between design parameters and flight characteristics
- Produced a formal technical report using LaTeX, documenting methodology, statistical analysis and conclusions
- Created supporting visualisations and 3D models to communicate experimental design

Fast Fourier Transform and Signal Analysis (1st Class) — pedroddvo/FFTPresentation

- Delivered a technical presentation explaining FFT algorithms and their applications in signal processing
- Explored the computational advantages of FFT algorithms compared with direct discrete Fourier transform methods
- Implemented FFT algorithms to analyse frequency components within sample datasets
- Communicated complex mathematical concepts to a technical audience through a formal presentation

Interests

Technical	Data Science, Mathematical Modelling, Open-source, Linux
Personal	Music production, Piano, Golf